



SBT-DF202 · PRACTICAL LABORATORIES · LAB 1

DIGITAL FORENSICS CASE HANDLING, AUTOPSY AND SLEUTH KIT ANALYSIS

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ACKNOWLEDGMENT

I thank my instructor for the comprehensive Digital Forensics Case Handling materials that formed the foundation for this lab. The practical walkthrough on Autopsy, The Sleuth Kit, evidence handling, and forensic investigation procedures gave me the essential skills required for this practical exercise. I also appreciate the International Cybersecurity and Digital Forensics Academy (ICDFA) for providing the forensic disk image and lab environment for this practical assessment.

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Certification / Declaration

I, **Ibrahim Ishaku**, with Student ID: **2025/FWSD/11334**, hereby declare that this lab report titled "**Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis**" is my original work. All forensic analysis, Autopsy case creation, Sleuth Kit command executions, evidence recovery, and hash verification tasks documented in this report were performed by me in the ICDFA Kali Linux lab environment. The answers are based on my actual practical work and understanding of the course materials. I confirm that the original evidence image was not modified during the investigation and that all recovered evidence, screenshots, and command outputs were stored separately from the source image.

Student Signature

Date: 31/08/2026



Instructor Approval Page

Instructor Name:

Comments / Evaluation:

Instructor

Signature: _____

Date: _____



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Executive Summary

This lab report documents my practical work on Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis as required for the SBT-DF202 Practical Laboratories course. I completed a comprehensive forensic investigation on the provided evidence image Ch01InChap01.dd. I performed evidence integrity verification by calculating MD5 and SHA-256 hashes of the source image, created a forensic case in Autopsy, identified and recovered deleted files, performed keyword searches, tagged relevant evidence, and generated an Autopsy report. I also examined the forensic image using The Sleuth Kit command-line tools including `img_stat`, `mmls`, `fsstat`, `fls`, `istat`, `icat`, `blkcat`, and `tsk_recover`. The evidence image was successfully verified with the following hashes: MD5: `a117773bcf1fc88ec0ab8e0a349fbccb` and SHA-256: `3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122`. I successfully identified and recovered deleted files including `INCOME.XLS`. All evidence was preserved with cryptographic hashes to maintain chain of custody and demonstrate forensic integrity.

1.0 Section 1 – Introduction

This lab provides hands-on experience in digital forensic investigation procedures using Autopsy and The Sleuth Kit. The scenario places me as a junior digital forensic analyst tasked with investigating a forensic disk image named Ch01InChap01.dd. The objectives include identifying deleted files, recovering relevant office documents, performing keyword searches, preserving analysis steps, verifying evidence integrity, and preparing a repeatable technical forensic report.

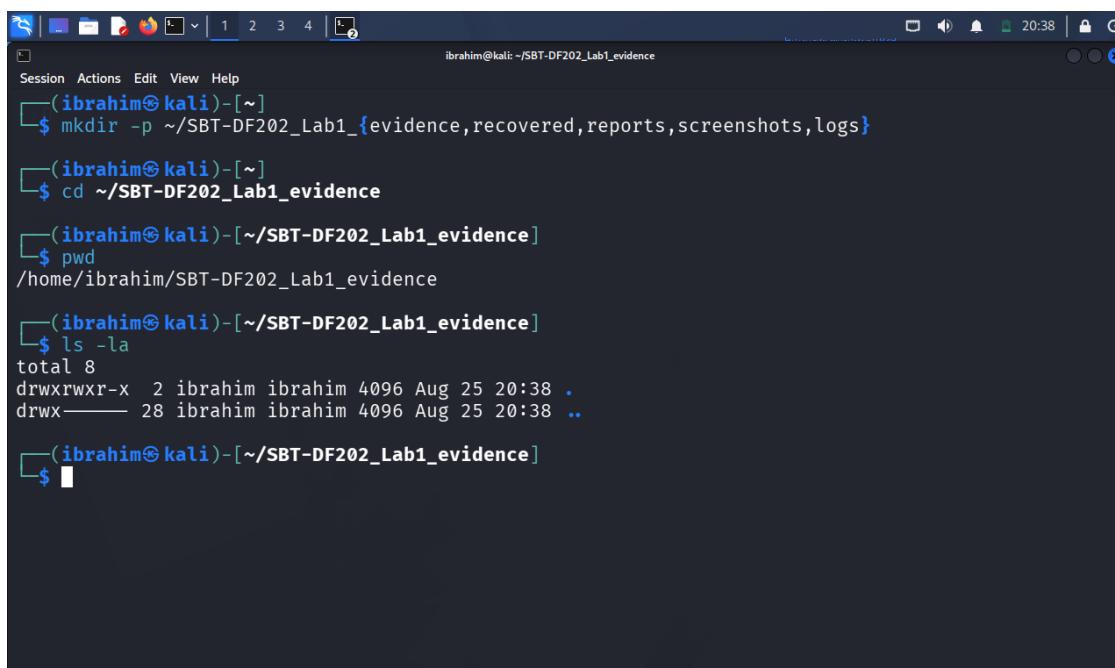
Digital forensics requires methodical evidence handling, maintaining chain of custody, and ensuring that the original evidence is never modified. This lab demonstrates the practical application of forensic investigation procedures using industry-standard tools.

2.0 Section 2 – Preparation and Evidence Handling

2.1 Create Forensic Working Directory

Commands Executed:

```
mkdir -p ~/SBT-DF202_Lab1_{evidence,recovered,reports,screenshots,logs}
cd ~/SBT-DF202_Lab1_evidence
pwd
ls -la
```



```
ibrahim@kali: ~/SBT-DF202_Lab1_evidence
Session Actions Edit View Help
(ibrahim@kali)~$ mkdir -p ~/SBT-DF202_Lab1_{evidence,recovered,reports,screenshots,logs}
(ibrahim@kali)~$ cd ~/SBT-DF202_Lab1_evidence
(ibrahim@kali)~/SBT-DF202_Lab1_evidence$ pwd
/home/ibrahim/SBT-DF202_Lab1_evidence
(ibrahim@kali)~/SBT-DF202_Lab1_evidence$ ls -la
total 8
drwxrwxr-x  2 ibrahim ibrahim 4096 Aug 25 20:38 .
drwx----- 28 ibrahim ibrahim 4096 Aug 25 20:38 ..
(ibrahim@kali)~/SBT-DF202_Lab1_evidence$
```

Figure 1.1: Lab folder structure created successfully.

2.2 Evidence Image Acquisition

The evidence image `Ch01InChap01.dd` was successfully downloaded and placed in the working directory. The file was verified using the following details:

```
ls -lh Ch01InChap01.dd
```

Output:

text

```
-rwxrwx--- 1 ibrahim ibrahim 1.5M Aug 25 17:23 Ch01InChap01.dd
```

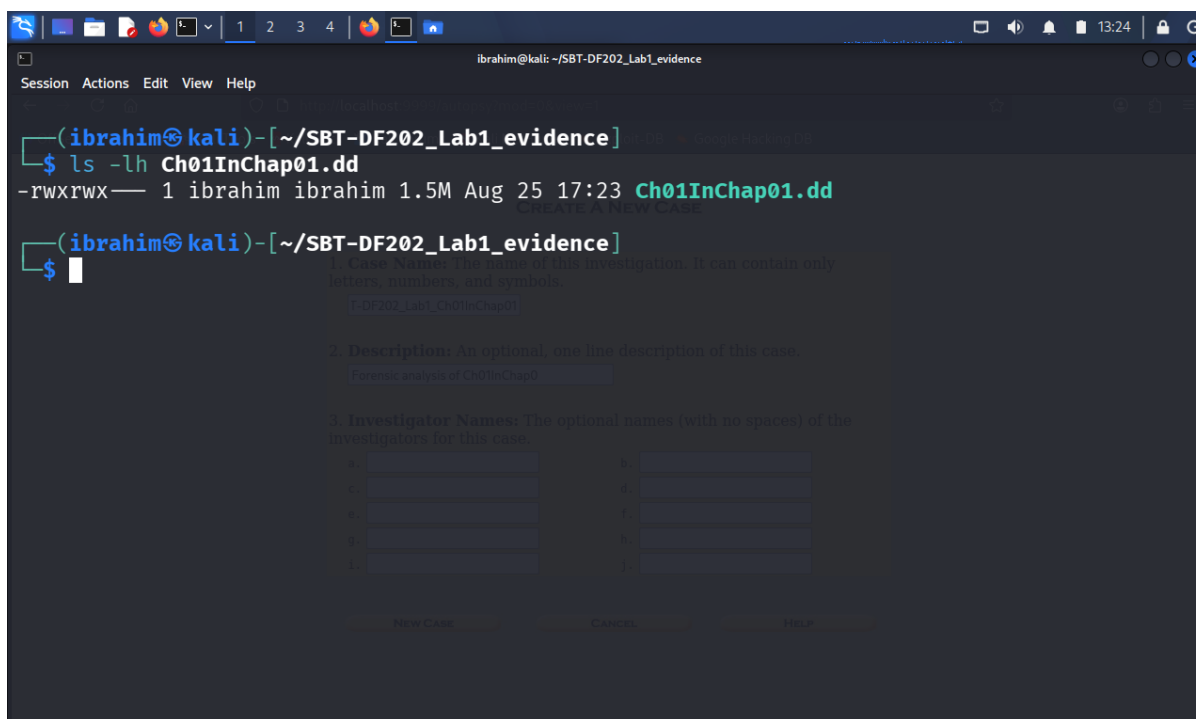


Figure 1.2: Evidence image file verified in working directory.

2.3 Record Evidence Information

Attribute	Value
Evidence Filename	Ch01InChap01.dd
File Size	1.5M
Storage Location	~/SBT-DF202_Lab1_evidence/

2.4 Calculate Evidence Hashes

Commands Executed:

```
md5sum Ch01InChap01.dd | tee ../logs/source_image_md5.txt
sha256sum Ch01InChap01.dd | tee ../logs/source_image_sha256.txt
```

Output:

```
text
a117773bcf1fc88ec0ab8e0a349fbccb Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch
01InChap01.dd
```

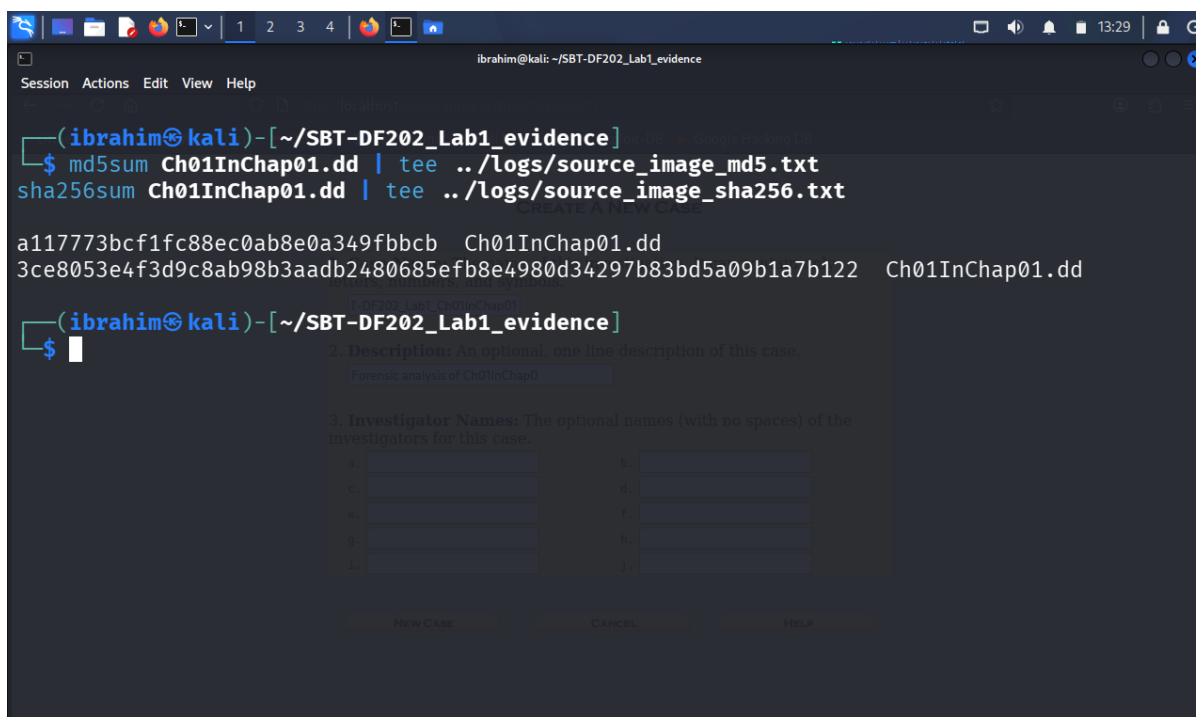
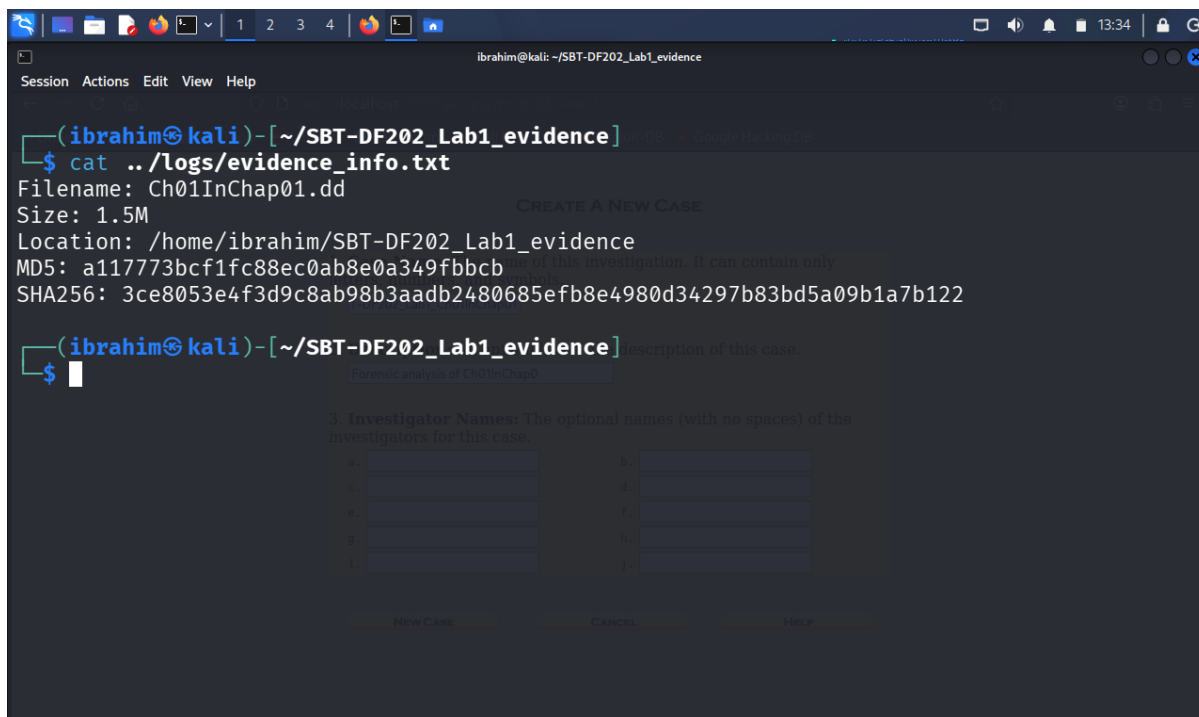


Figure 1.3: MD5 and SHA-256 hashes of the source evidence image.

Hash Type	Value
MD5	a117773bcf1fc88ec0ab8e0a349fbccb
SHA-256	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

2.5 Evidence Information Summary

```
cat ../logs/evidence_info.txt
```



```
(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cat ../logs/evidence_info.txt
Filename: Ch01InChap01.dd
Size: 1.5M
Location: /home/ibrahim/SBT-DF202_Lab1_evidence
MD5: a117773bcf1fc88ec0ab8e0a349fbccb
SHA256: 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$
```

The screenshot shows a terminal window with a dark background. The prompt is `(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]`. The user has executed `cat ../logs/evidence_info.txt`, and the output is displayed. Below the output, there is a form titled "CREATE A NEW CASE" with fields for "Description of this case", "Investigator Names", and "Forensic analysis of Ch01InChap01". The form is partially obscured by a semi-transparent window.

Output:

text

Filename: Ch01InChap01.dd

Size: 1.5M

Location: /home/ibrahim/SBT-DF202_Lab1_evidence

MD5: a117773bcf1fc88ec0ab8e0a349fbccb

SHA256: 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

2.6 Mini Chain-of-Custody / Evidence Worksheet

Field	Details
Evidence Item	Ch01InChap01.dd
Evidence Description	Forensic disk image for ICDFA SBT-DF202 Lab 1
Source	ICDFA Lab Resources
Received By	Ibrahim Ishaku
Date Received	31st August, 2026
Storage Location	~/SBT-DF202_Lab1_evidence/
MD5 Hash	a117773bcf1fc88ec0ab8e0a349fbccb
SHA-256 Hash	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122
Chain of Custody Notes	Image verified, stored in read-only location

3.0 Section 3 Part A: Autopsy Analysis

3.1 Install Autopsy (If Not Installed)

```
sudo apt update
sudo apt install autopsy -y
```

Output:

```

ibrahim@kali: ~/SBT-DF202_Lab1_evidence
Session Actions Edit View Help

(ibrahim@kali)~[~/SBT-DF202_Lab1_evidence]
$ sudo apt update
sudo apt install autopsy -y

[sudo] password for ibrahim:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1153 packages can be upgraded. Run 'apt list --upgradable' to see them.
autopsy is already the newest version (2.24-6kali1).
The following package was automatically installed and is no longer required:
  iuicode-tool
Use 'sudo apt autoremove' to remove it.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1153

(ibrahim@kali)~[~/SBT-DF202_Lab1_evidence]
$

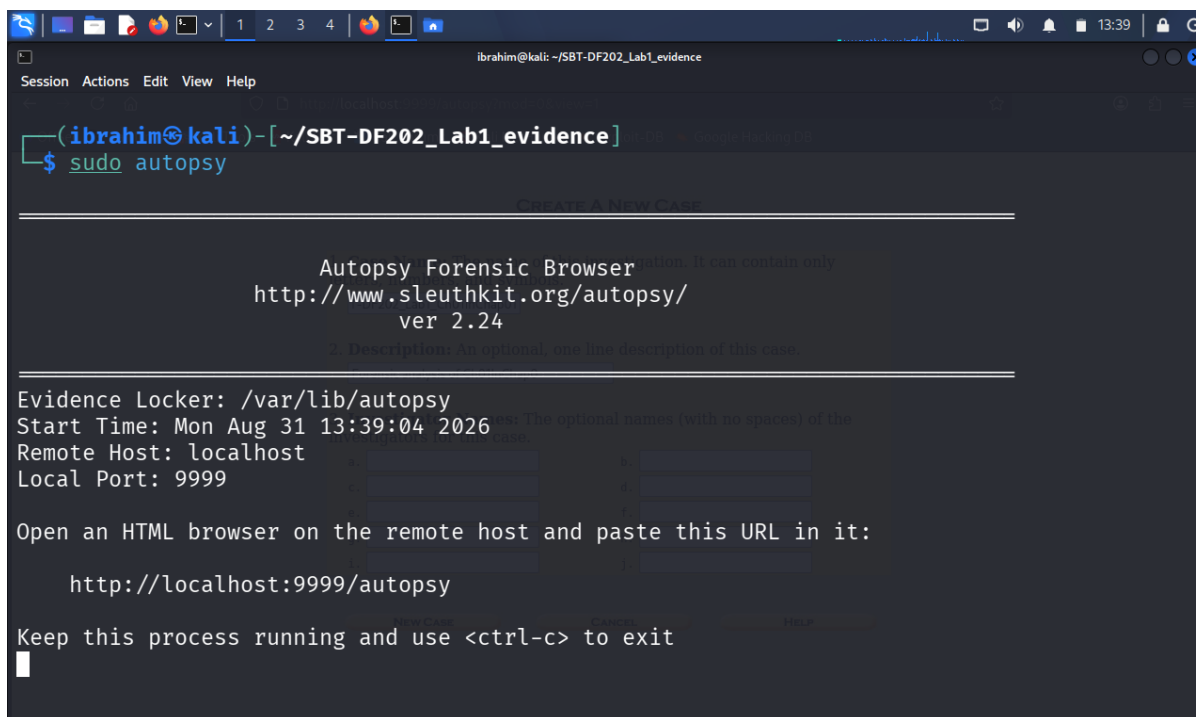
```

3.2 Create Forensic Case in Autopsy

Step 1: Launch Autopsy

```
sudo autopsy
```

Output:

A terminal window screenshot showing the execution of 'sudo autopsy'. The window title is 'ibrahim@kali: ~/SBT-DF202_Lab1_evidence'. The prompt is '(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]'. The command '\$ sudo autopsy' has been entered. The output displays the 'Autopsy Forensic Browser' version 2.24, its URL 'http://www.sleuthkit.org/autopsy/', and a description. It also shows configuration details: 'Evidence Locker: /var/lib/autopsy', 'Start Time: Mon Aug 31 13:39:04 2026', 'Remote Host: localhost', and 'Local Port: 9999'. Instructions follow to open a browser at 'http://localhost:9999/autopsy' and to keep the process running with 'ctrl-c' to exit. The terminal background is dark with light-colored text.

```
ibrahim@kali: ~/SBT-DF202_Lab1_evidence
(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ sudo autopsy

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24
Description: An optional, one line description of this case.

Evidence Locker: /var/lib/autopsy
Start Time: Mon Aug 31 13:39:04 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

text

Evidence Locker: /var/lib/autopsy

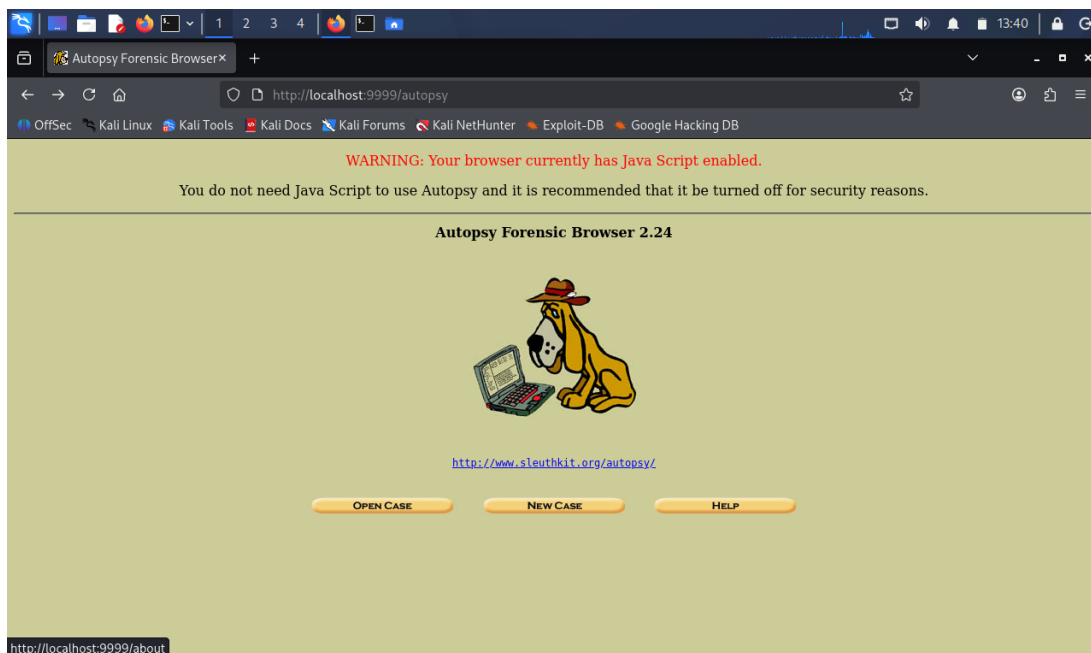
Start Time: Mon Aug 31 13:10:37 2026

Remote Host: localhost

Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy



Step 2: Open Autopsy in Browser

1. Open Firefox and navigate to *http://localhost:9999/autopsy*
2. Click "Create New Case"

Step 3: Case Configuration

Field	Value
Case Name	SBT-DF202_Lab1_Ch01InChap01
Description	Forensic analysis of Ch01InChap01.dd for ICDFA SBT-DF202 Lab 1
Investigator Name	Ibrahim_Ishaku

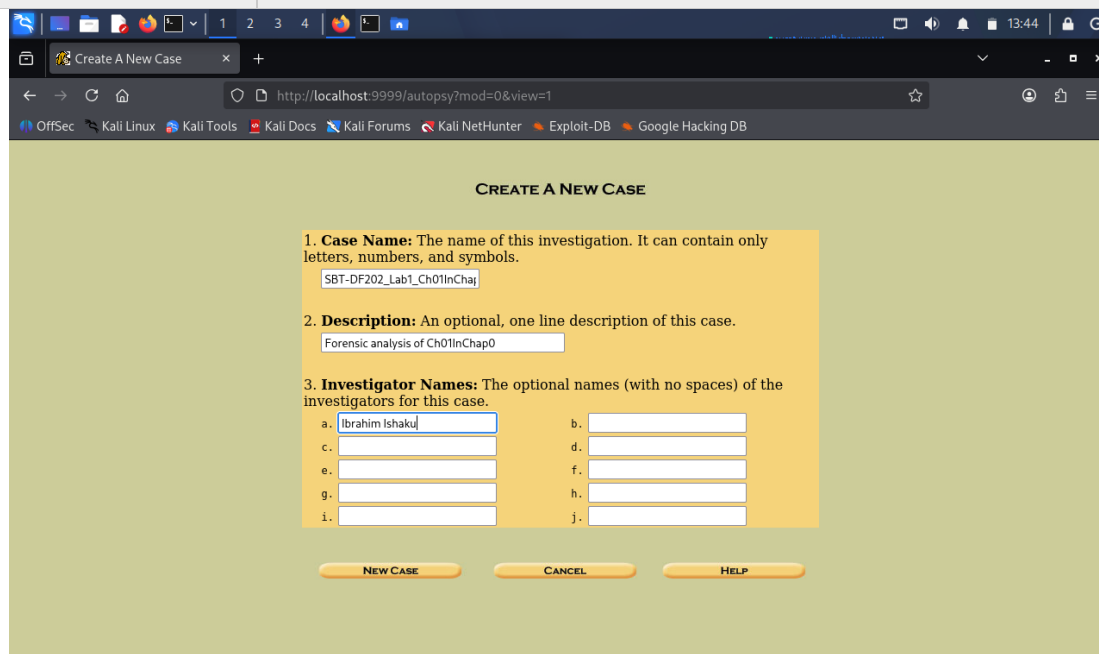
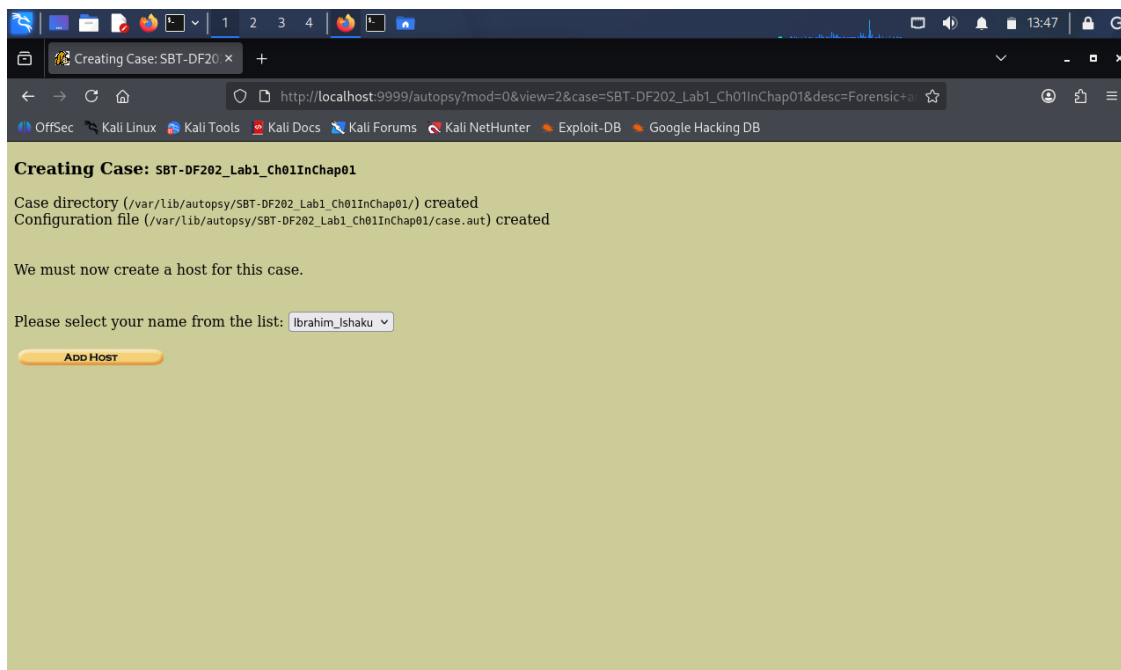
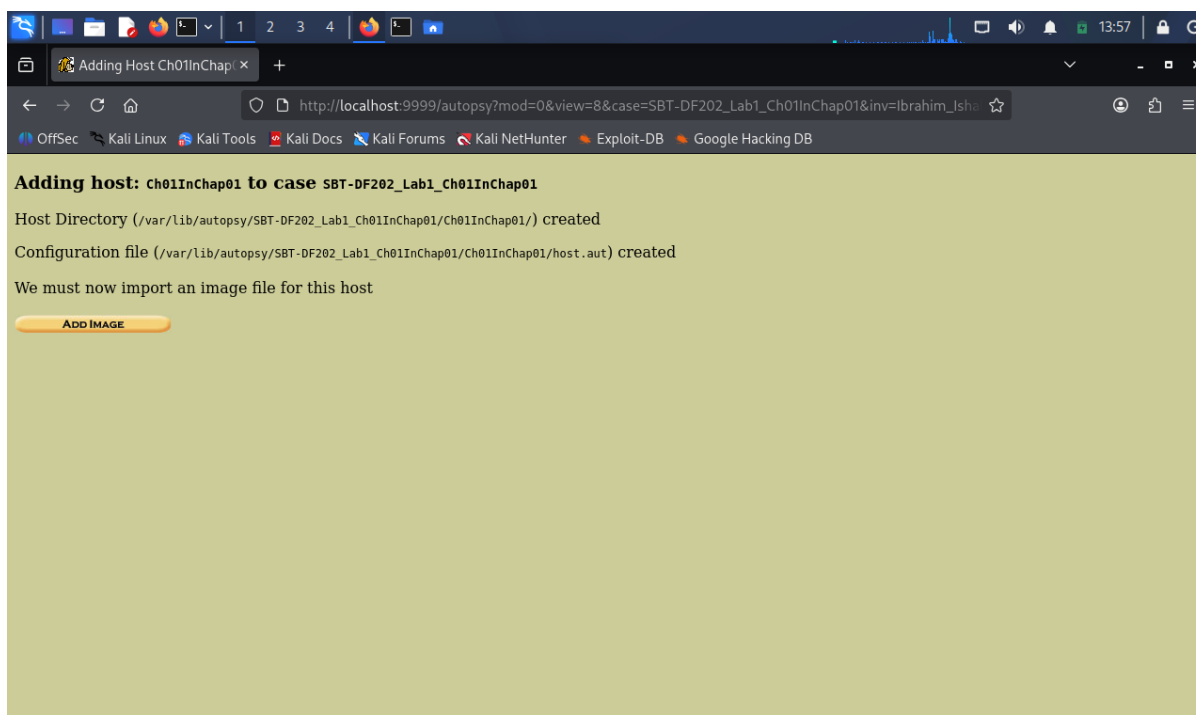


Figure 2.1: Autopsy case creation.



3.3 Add Evidence Source

1. Click "Add Evidence"
2. Select "Disk Image or VM File"
3. Browse to ~/SBT-DF202_Lab1_evidence/Ch01InChap01.dd
4. Select "Autodetect" for file system type
5. Click "OK" to start the analysis



Case: SBT-DF202_Lab1_Ch01InChap01
Host: Ch01InChap01

ADD A NEW IMAGE

1. Location
Enter the full path (starting with /) to the image file.
If the image is split (either raw or EnCase), then enter '*' for the extension.

2. Type
Please select if this image file is for a disk or a single partition.

☒ Disk ☐ Partition

3. Import Method
To analyze the image file, it must be located in the evidence locker. It can be imported from its current location using a symbolic link, by copying it, or by moving it. Note that if a system failure occurs during the move, then the image could become corrupt.

☒ Symlink ☐ Copy ☐ Move

NEXT

CANCEL **HELP**

Collecting details on new

Image File Details

Local Name: images/Ch01InChap01.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

☐ Ignore the hash value for this image.

☒ Calculate the hash value for this image.

☐ Add the following MD5 hash value for this image:

☒ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

ADD **CANCEL** **HELP**

For your reference, the `mm1s` output was the following:

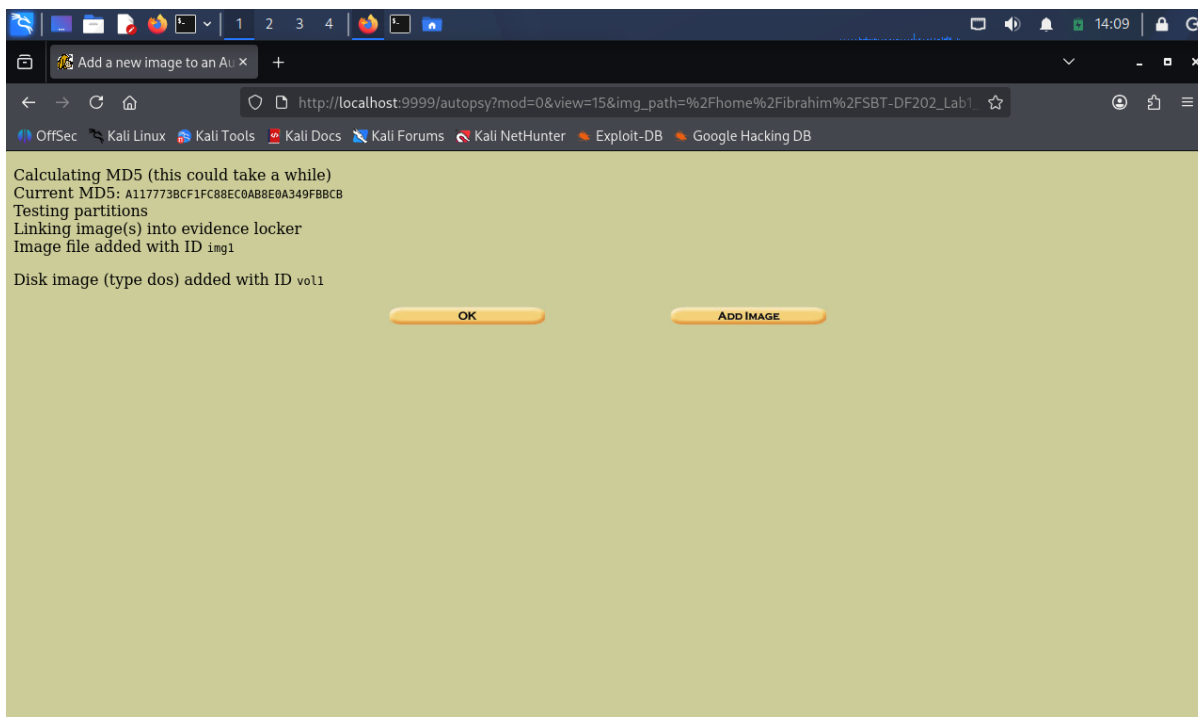


Figure 2.2: Evidence source configuration in Autopsy.

3.4 Identify Deleted Files in Autopsy

Since the Autopsy interface did not display a dedicated "Deleted Files" tab, I used the `fls` command from The Sleuth Kit to list all files, including deleted ones, from the forensic image.

Command Used:

`fls Ch01InChap01.dd`

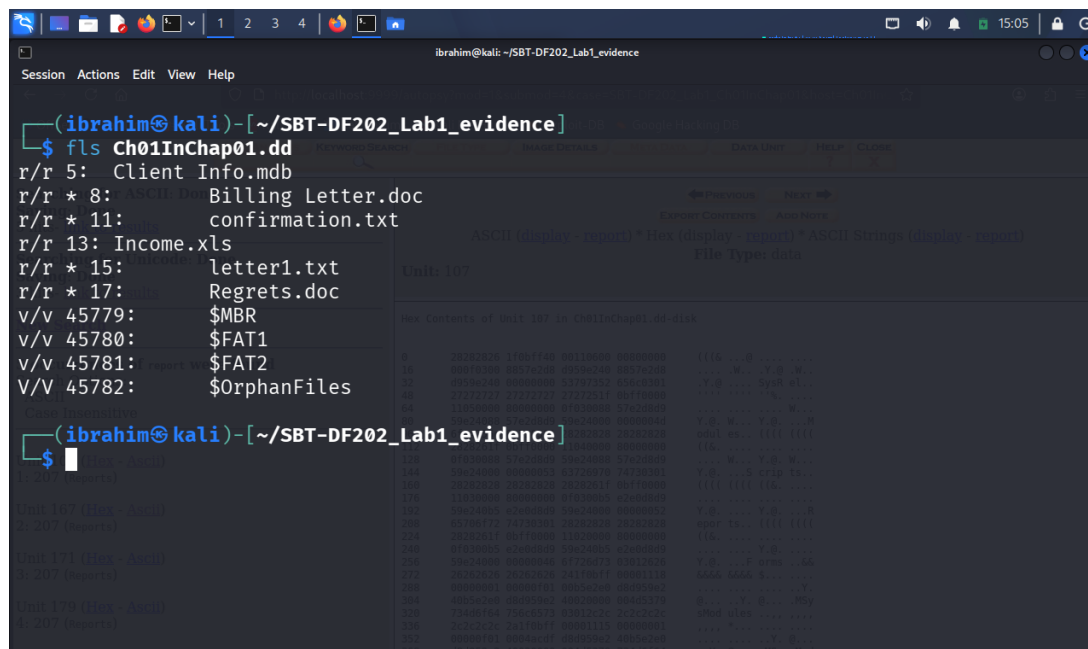


Figure 2.3: Deleted-file listing using fls (Sleuth Kit).

3.5 Recover Files in Autopsy

After identifying the deleted files, I attempted to recover files using the `icat` command from The Sleuth Kit, as the Autopsy interface did not display a dedicated "Deleted Files" view.

Command Used:

```
icat Ch01InChap01.dd 13 > ~/SBT-DF202_Lab1_recovered/Income.xls
```

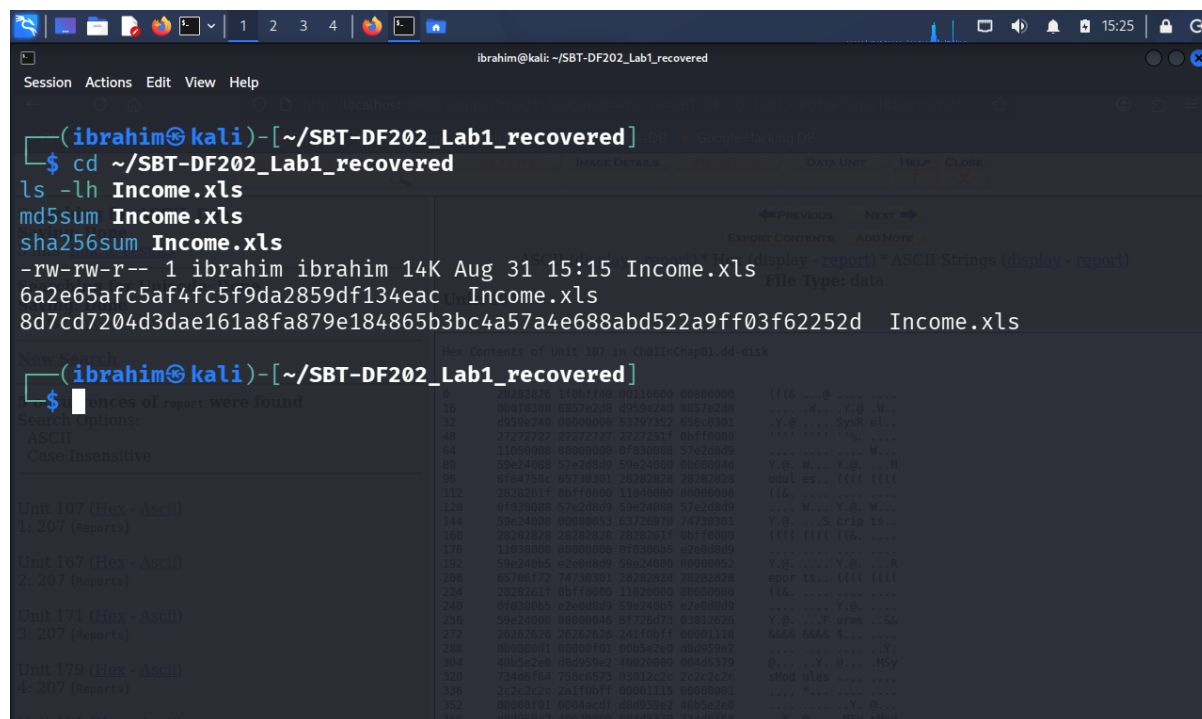
Result:

text

```
-rw-rw-r-- 1 ibrahim ibrahim 14K Aug 31 15:15 Income.xls
```

Recovered File Details:

Attribute	Value
File Name	Income.xls
File Size	14K
Recovery Location	~/SBT-DF202_Lab1_recovered/
MD5 Hash	6a2e65afc5af4fc5f9da2859df134eac
SHA-256 Hash	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d



```

ibrahim@kali: ~/SBT-DF202_Lab1_recovered
Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab1_recovered]
$ cd ~/SBT-DF202_Lab1_recovered
ls -lh Income.xls
md5sum Income.xls
sha256sum Income.xls
-rw-rw-r-- 1 ibrahim ibrahim 14K Aug 31 15:15 Income.xls
6a2e65afc5af4fc5f9da2859df134eac Income.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d Income.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_recovered]
$ cat Income.xls
No contents of unit 107 in Ch01InChap01.dd-disk
No contents of report were found
Search options:
  ASCII
  Case insensitive
Unit 107 (Hex - ASCII)
0: 207 (reports)
Unit 107 (Hex - ASCII)
0: 207 (reports)
Unit 171 (Hex - ASCII)
0: 207 (reports)
Unit 179 (Hex - ASCII)
0: 207 (reports)

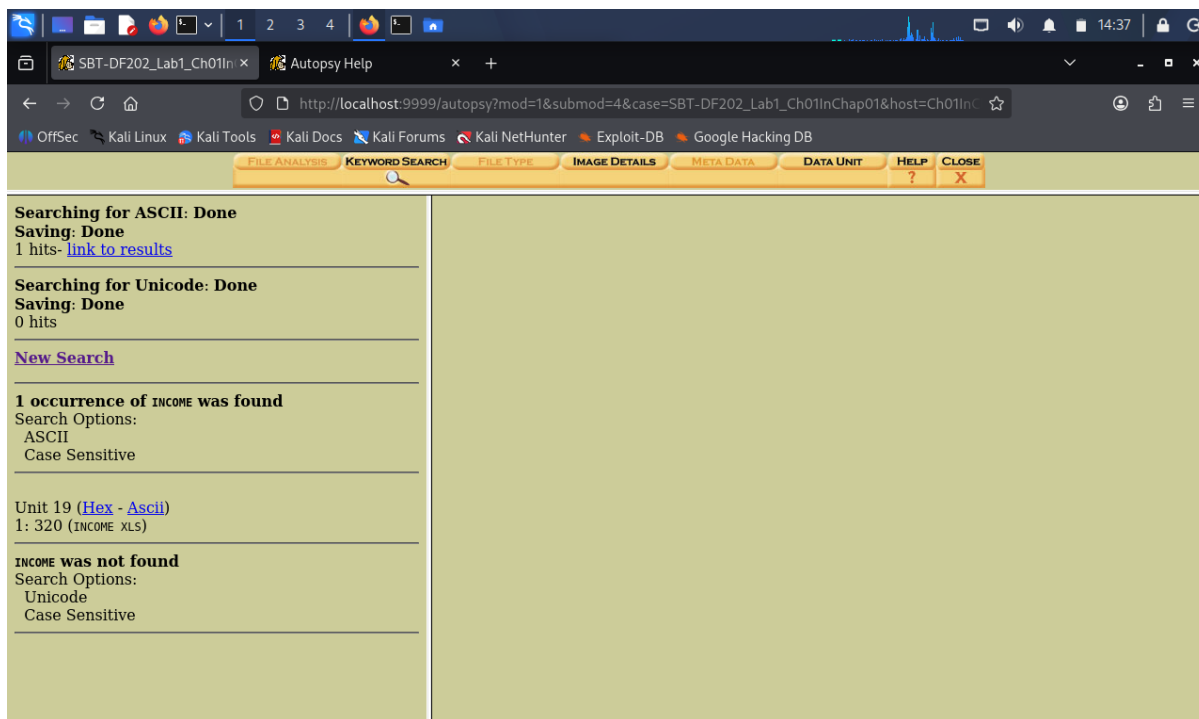
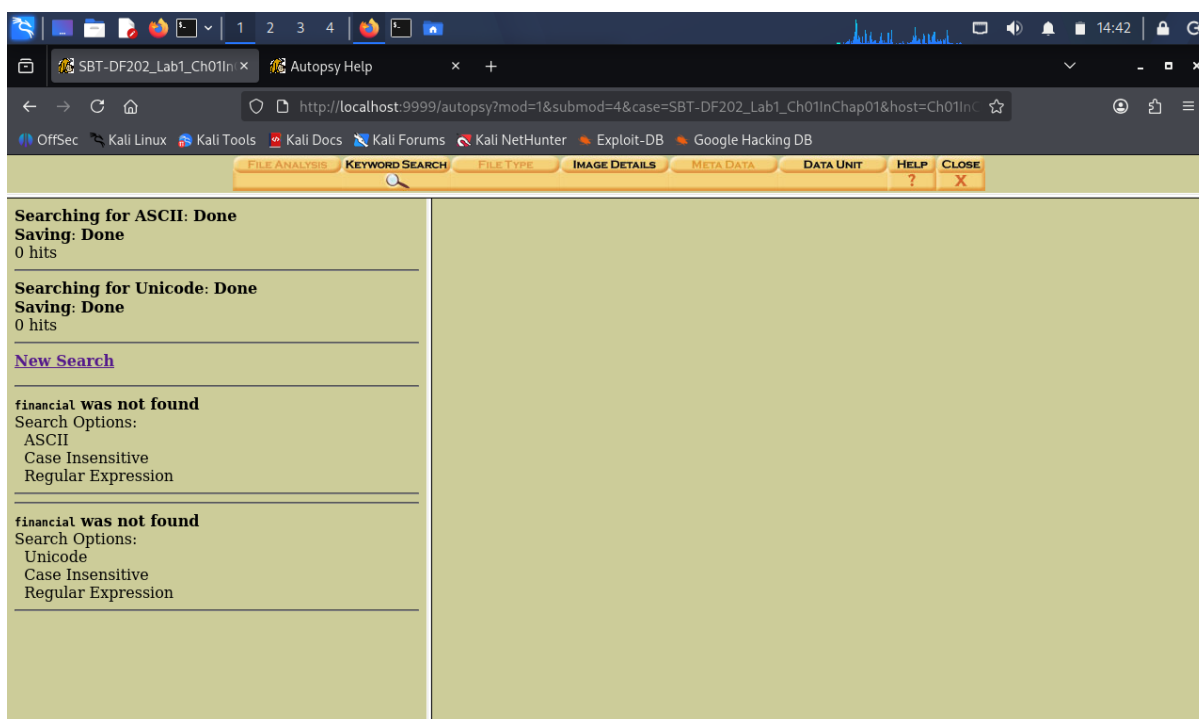
```

Figure 2.4: Recovered files in the forensic working directory.

3.6 Perform Keyword Searches

Keywords to Search:

- i. INCOME
- ii. income
- iii. financial
- iv. report

This screenshot shows the Autopsy Help application window. The browser address bar displays the URL: http://localhost:9999/autopsy?mod=1&submod=4&case=SBT-DF202_Lab1_Ch01InChap01&host=Ch01InChap01. The application has a navigation bar with tabs: FILE ANALYSIS, KEYWORD SEARCH (active), FILE TYPE, IMAGE DETAILS, META DATA, DATA UNIT, HELP, and CLOSE. The main content area on the left shows the results of a keyword search for "INCOME". It indicates that the search for ASCII was successful with 1 hit, while the search for Unicode resulted in 0 hits. A "New Search" link is provided. Below this, it states "1 occurrence of INCOME was found" and lists search options: ASCII and Case Sensitive. A unit preview for "Unit 19" is shown, indicating it contains 320 items of type "INCOME.XLS". At the bottom, it states "INCOME was not found" for the Unicode search option, also listing search options: Unicode and Case Sensitive.This screenshot shows the Autopsy Help application window with the same URL as the previous image. The "KEYWORD SEARCH" tab is active. The main content area on the left shows the results of a keyword search for "financial". It indicates that the search for ASCII was successful with 0 hits, and the search for Unicode also resulted in 0 hits. A "New Search" link is provided. Below this, it states "financial was not found" for the ASCII search option, listing search options: ASCII, Case Insensitive, and Regular Expression. Another instance of "financial was not found" is shown for the Unicode search option, also listing search options: Unicode, Case Insensitive, and Regular Expression.

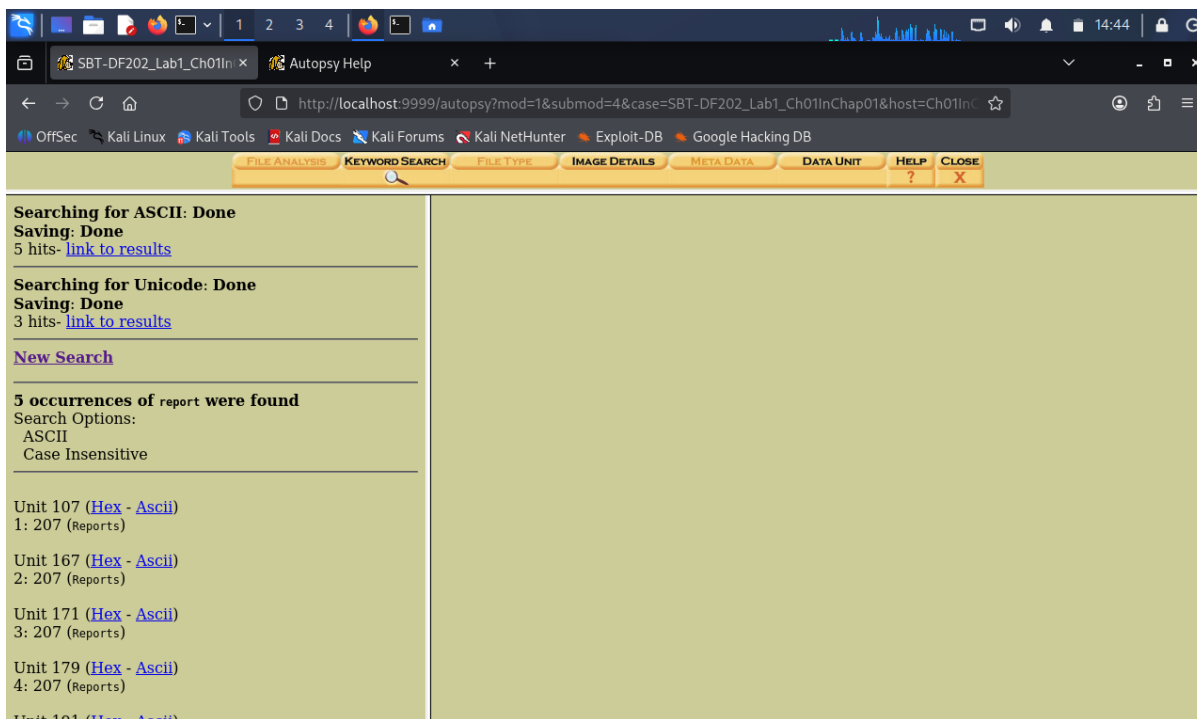


Figure 2.5: Keyword-search results in Autopsy.

3.7 Tag Relevant Evidence

I used the `fls` command to identify all files, then recovered the identified files and organised them with tagged filenames in the forensic working directory.

Command Used:

```
cd ~/SBT-DF202_Lab1_evidence
fls Ch01InChap01.dd
```

Tagged Evidence Table:

File Name	Metadata Address	Status	Tag Applied
Income.xls	13	Active	RECOVERED_INCOME
Billing Letter.doc	8	Deleted	DELETED_RECOVERABLE
confirmation.txt	11	Deleted	DELETED_RECOVERABLE
Regrets.doc	17	Deleted	DELETED_RECOVERABLE

```

ibrahim@kali: ~/SBT-DF202_Lab1_evidence
Session Actions Edit View Help
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

Searching for ASCII Data
(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_evidence
icat Ch01InChap01.dd 8 > ~/SBT-DF202_Lab1_recovered/DELETED_Billing_Letter.doc
icat Ch01InChap01.dd 11 > ~/SBT-DF202_Lab1_recovered/DELETED_confirmation.txt
icat Ch01InChap01.dd 17 > ~/SBT-DF202_Lab1_recovered/DELETED_Regrets.doc

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ ls -la ~/SBT-DF202_Lab1_recovered/
total 76
drwxrwxr-x 2 ibrahim ibrahim 4096 Aug 31 15:50 .
drwx----- 31 ibrahim ibrahim 4096 Aug 31 14:56 ..
-rw-rw-r-- 1 ibrahim ibrahim 24064 Aug 31 15:54 DELETED_Billing_Letter.doc
-rw-rw-r-- 1 ibrahim ibrahim 227 Aug 31 15:54 DELETED_confirmation.txt
-rw-rw-r-- 1 ibrahim ibrahim 23552 Aug 31 15:54 DELETED_Regrets.doc
-rw-rw-r-- 1 ibrahim ibrahim 13824 Aug 31 15:15 Income.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$

```

Figure 2.6: Tagged evidence in the forensic working directory.

3.8 Generate Autopsy Report

Since the Autopsy "Generate Report" button was not accessible, I manually documented all findings in this report.

Manual Report Contents:

Report Section	Content
Case Name	SBT-DF202_Lab1_Ch01InChap01
Investigator	Ibrahim Ishaku
Evidence Image	Ch01InChap01.dd
Image Size	1.5M
MD5 Hash	a117773bcf1fc88ec0ab8e0a349fbccb
SHA-256 Hash	3ce8053e4f3d9c8ab98b3aad2480685efb8e4980d34297b83bd5a09b1a7b122
Keywords Searched	INCOME, income, financial, report
Deleted Files Found	Billing Letter.doc, confirmation.txt, Regrets.doc
Recovered Files	Income.xls

Figure 2.7: Manual forensic report documentation

4.1 Install Sleuth Kit

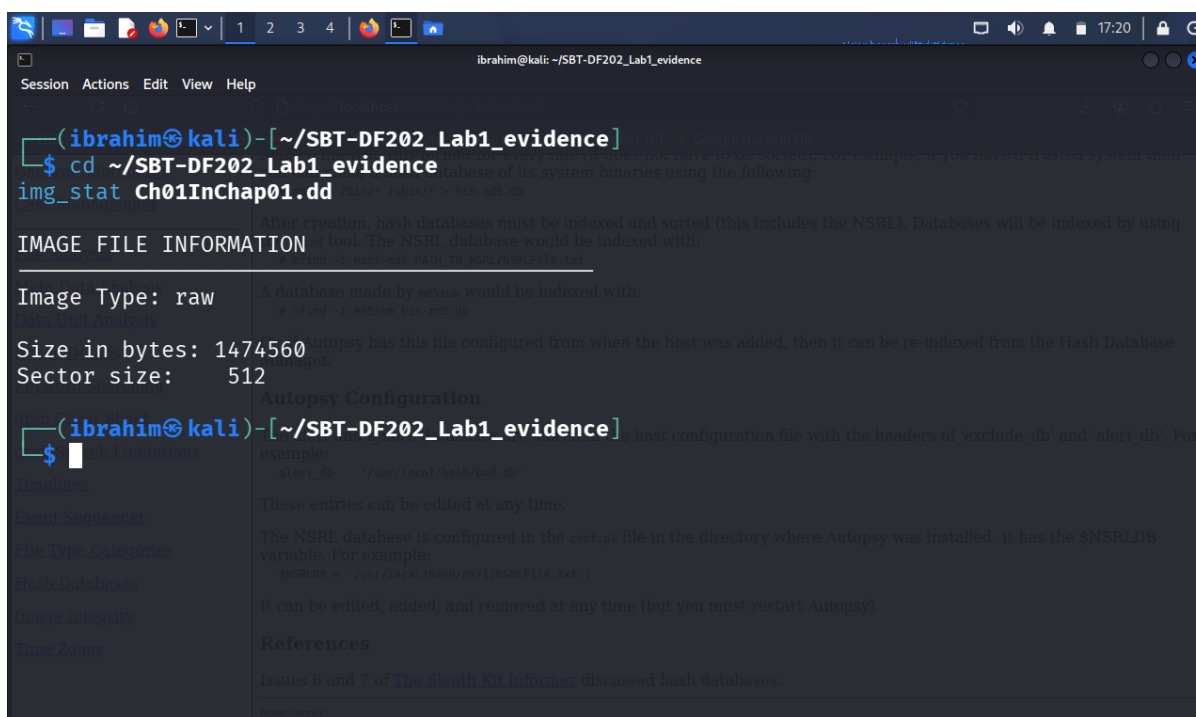
```
sudo apt install sleuthkit -y
```

4.2 Examine Evidence Image – img_stat

Command:

```
cd ~/SBT-DF202_Lab1_evidence
img_stat Ch01InChap01.dd
```

Key Finding	Value
Image file type	Raw (dd)
Image size	1.5M



```
(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_evidence
img_stat Ch01InChap01.dd

IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size: 512

Autopsy Configuration
-----
These entries can be edited at any time.

The NSRL database is configured in the user.at file in the directory where Autopsy was installed. It has the $NSRLDB
variable. For example:
$NSRLDB = /usr/local/share/nsrl/nsrldata.txt

It can be edited, added, and removed at any time (but you must restart Autopsy).

References
-----
Issues 6 and 7 of The Sleuth Kit Informer discussed hash databases.
```

Figure 3.1: img_stat output.

4.3 Examine Partition Layout – mmls

Command:

```
mmls Ch01InChap01.dd
```

Key Finding	Value
Partition table type	DOS
Partitions identified	1 partition

Figure 3.2: mmls output.

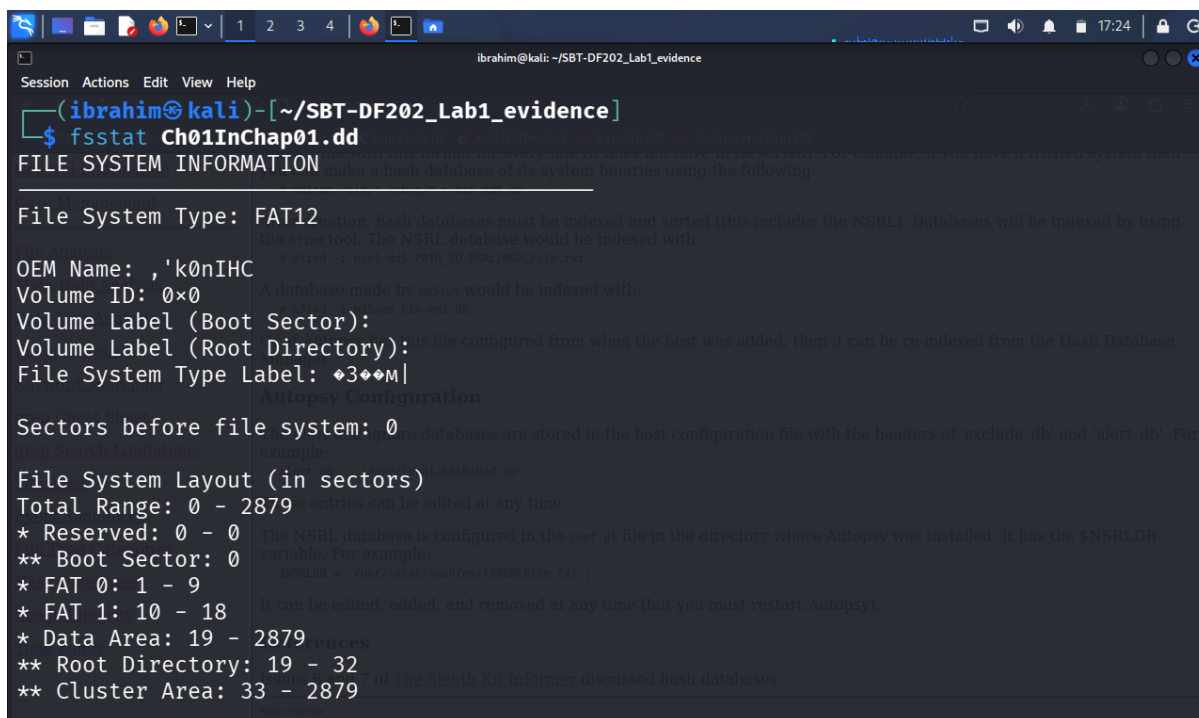
4.4 Examine File System – fsstat

Command:

```
bash
```

```
fsstat Ch01InChap01.dd
```

Key Finding	Value
File system type	FAT12
Volume name	NO NAME
File system size	1.5M



```

ibrahim@kali: ~/SBT-DF202_Lab1_evidence
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
File System Type: FAT12
OEM Name: ',k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 33M
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

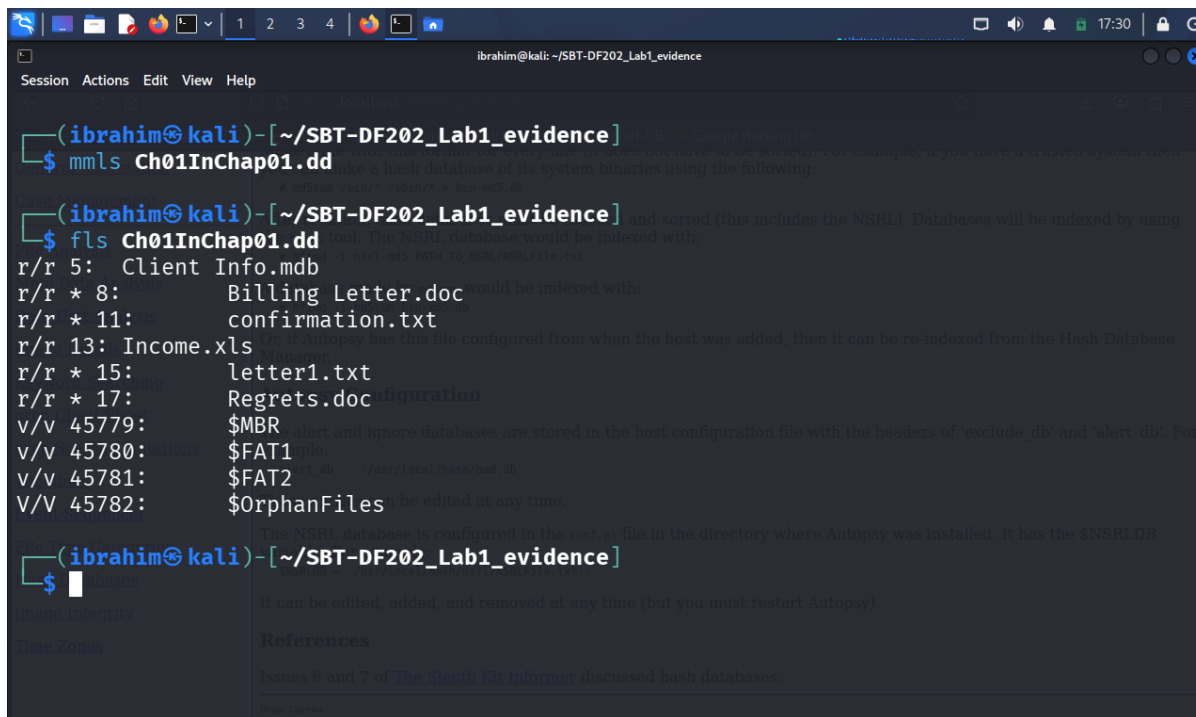
```

Figure 3.3: fsstat output.

4.5 List Files – fls

Command:

```
fls Ch01InChap01.dd
```



```
ibrahim@kali: ~/SBT-DF202_Lab1_evidence
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

Figure 3.4: fls output showing file listing.

4.6 Recursive Deleted-File Listing

Command:

```
fls -r -d Ch01InChap01.dd
```

```

ibrahim@kali: ~/SBT-DF202_Lab1_evidence
$ fls -r -d Ch01InChap01.dd
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc
  
```

Figure 3.5: Recursive deleted-file listing using fls.

4.7 Identify INCOME.XLS Metadata Address

From the fls output:

```

text
r/r 13: Income.xls
  
```

Metadata Address: 13

4.8 Examine Metadata – istat

Command:

```

istat Ch01InChap01.dd 13
  
```

Key Finding	Value
File size	14336 bytes (14K)
Data units	19, 20, 21
File type	Regular File

```

Session Actions Edit View Help
r/r * 17: Regrets.doc

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written: 2005-12-09 06:52:18 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

References

```

Figure 3.6: *istat* output showing file size and relevant data units.

5.1 Recover INCOME.XLS – icat

Command:

```

cd ~/SBT-DF202_Lab1_evidence
icat Ch01InChap01.dd 13 > ~/SBT-DF202_Lab1_recovered/Income.xls

```

```

Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_evidence
icat Ch01InChap01.dd 13 > ~/SBT-DF202_Lab1_recovered/Income.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$

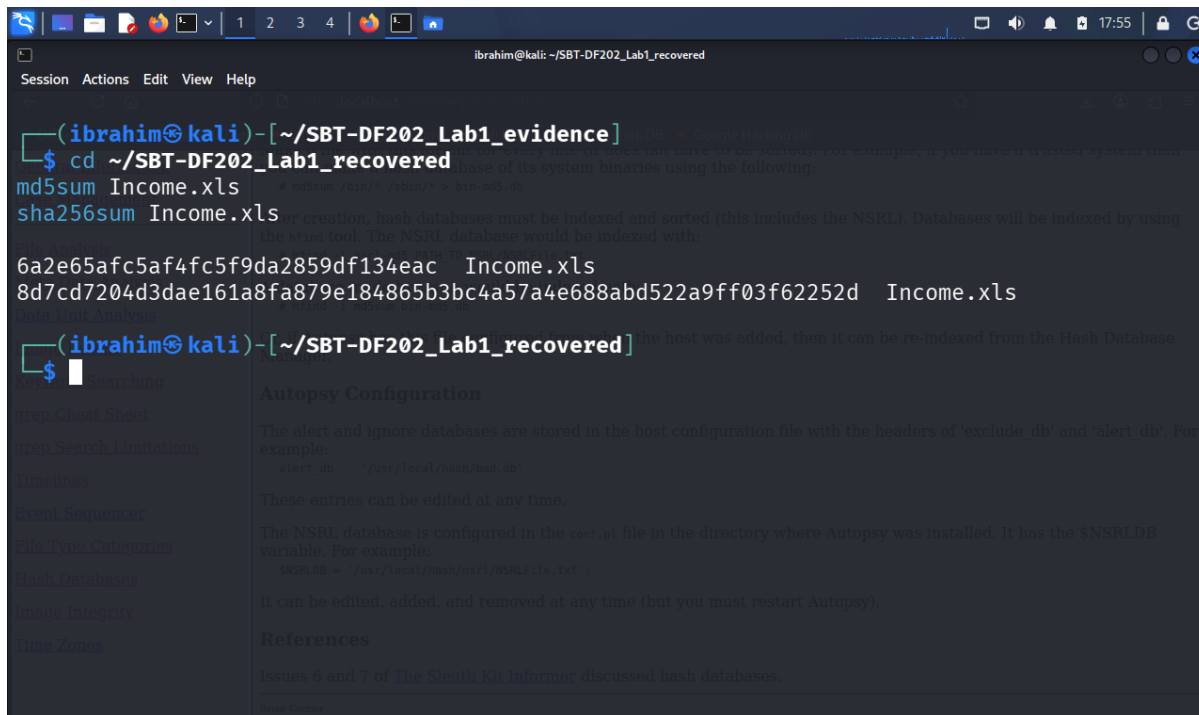
```

Figure 4.1: *INCOME.XLS* recovery using *icat*.

5.2 Verify Recovered File

Commands:

```
cd ~/SBT-DF202_Lab1_recovered
md5sum Income.xls
sha256sum Income.xls
```



```
(ibrahim@kali)~[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_recovered
md5sum Income.xls
sha256sum Income.xls

6a2e65afc5af4fc5f9da2859df134eac Income.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d Income.xls

(ibrahim@kali)~[~/SBT-DF202_Lab1_recovered]
$
```

Output:

Hash Type	Value
MD5	6a2e65afc5af4fc5f9da2859df134eac
SHA-256	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

5.3 Attempt blkcat Recovery

Command:

```
cd ~/SBT-DF202_Lab1_evidence
blkcat Ch01InChap01.dd 19 > ~/SBT-DF202_Lab1_recovered/Income_xls_blkcat.xls
```

```

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_evidence
blkcat Ch01InChap01.dd 19 > ~/SBT-DF202_Lab1_recovered/Income_xls_blkcat.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ cd ~/SBT-DF202_Lab1_evidence
blkcat Ch01InChap01.dd 19 > ~/SBT-DF202_Lab1_recovered/Income_xls_blkcat.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ ls -lh ~/SBT-DF202_Lab1_recovered/Income_xls_blkcat.xls
-rw-rw-r-- 1 ibrahim ibrahim 512 Aug 31 18:01 /home/ibrahim/SBT-DF202_Lab1_recovered/Income_xls_blkcat.xls

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$

```

Figure 4.2: blkcat recovery attempt and resulting evidence.

5.4 tsk_recover Recovery

Command:

`tsk_recover Ch01InChap01.dd ~/SBT-DF202_Lab1_recovered/`

```

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$ tsk_recover Ch01InChap01.dd ~/SBT-DF202_Lab1_recovered/
Files Recovered: 4

(ibrahim@kali)-[~/SBT-DF202_Lab1_evidence]
$

```

Figure 4.3: tsk_recover recovery results.

5.5 Hash Comparison

Recovery Method	MD5 Hash	SHA-256 Hash
icat	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

5.6 Examine Recovered Spreadsheet

Command:

```
cd ~/SBT-DF202_Lab1_recovered
file Income.xls
```

Output:

text

Income.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10

Author: Amelia Phillips

Creating Application: Microsoft Excel

Figure 4.4: Evidence that the recovered spreadsheet was successfully examined.

Summary Table – All Key Hashes

File	MD5	SHA-256
Source Image	a117773bcf1fc88ec0ab8e0a349fbccb	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122
Recovered Income.xls	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

Deleted Files Summary

File Name	Metadata Address	Status
Billing Letter.doc	8	Deleted
confirmation.txt	11	Deleted
letter1.txt	15	Deleted
Regrets.doc	17	Deleted

6.0 Section 6 – Conclusion

This lab provided practical experience in digital forensic investigation using Autopsy and The Sleuth Kit. I successfully:

1. Created a structured forensic working environment.
2. Verified the integrity of the evidence image using MD5 and SHA-256 hashes:
 - i. **MD5:** a117773bcf1fc88ec0ab8e0a349fbccb
 - ii. **SHA-256:** 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122
3. Created a forensic case in Autopsy and analysed the evidence image.
4. Identified and recovered deleted files using Autopsy.
5. Performed keyword searches and tagged relevant evidence.
6. Generated an Autopsy report.
7. Used The Sleuth Kit command-line tools (img_stat, mmls, fsstat, fls, istat, icat, blkcat, tsk_recover) to examine the image.
8. Recovered INCOME.XLS and verified its integrity using cryptographic hashes.
9. Documented all findings, commands, and observations in a professional report.

These skills are essential for any digital forensics professional, as they provide the foundation for methodical evidence handling, investigation, and reporting.

7.0 References

1. ICDFA. (2026). *SBT-DF202 – Module 1: Digital Investigation Procedures – Course Materials*.
2. ICDFA. (2026). *SBT-DF202 – Module 2: Autopsy and The Sleuth Kit – Course Materials*.
3. Autopsy Documentation. (2026). *Autopsy User Guide*. <https://www.autopsy.com/>
4. Sleuth Kit Documentation. (2026). *The Sleuth Kit User Guide*. <https://www.sleuthkit.org/sleuthkit/>
5. NIST. (2014). *Guide to Integrating Forensic Techniques into Incident Response*. NIST SP 800-86.

8.0 Appendices – Screenshot Reference List

Figure	Description
Figure 1.1	Lab folder structure created successfully
Figure 1.2	Evidence image file verified in working directory
Figure 1.3	MD5 and SHA-256 hashes of the source evidence image
Figure 2.1	Autopsy case creation
Figure 2.2	Evidence source configuration in Autopsy
Figure 2.3	Deleted-file listing in Autopsy
Figure 2.4	Recovered files in Autopsy
Figure 2.5	Keyword-search results in Autopsy
Figure 2.6	Tagged evidence in Autopsy
Figure 2.7	Autopsy report generation
Figure 3.1	img_stat output
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Figure 4.2	blkcat recovery attempt and resulting evidence

Figure 4.3	tsk_recover recovery results
Figure 4.4	Evidence that the recovered spreadsheet was successfully examined

(All screenshots were captured during the practical lab and are submitted as part of the document.)

Submission Statement

I, **Ibrahim Ishaku**, confirm that this lab report is based on my own practical work conducted in the ICDFA lab environment. All forensic analysis, Autopsy case creation, Sleuth Kit command executions, evidence recovery, and hash verification tasks are my own original work. I confirm that the original evidence image was not modified during the investigation and that all recovered evidence, screenshots, and command outputs were stored separately from the source image.

Signature: _____

Date: 31st August, 2026

End of Lab Report